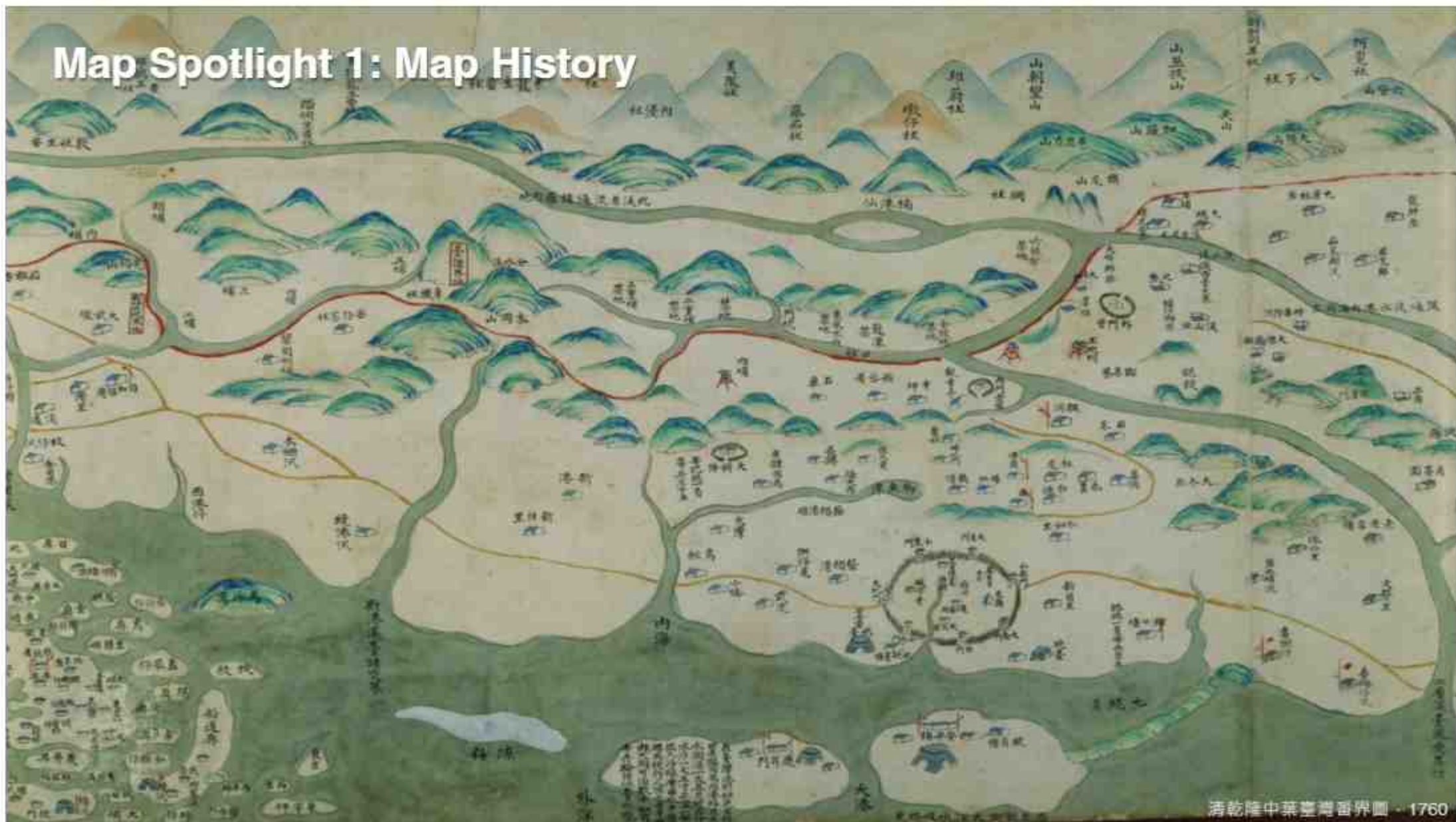


Map Spotlight 1: Map History

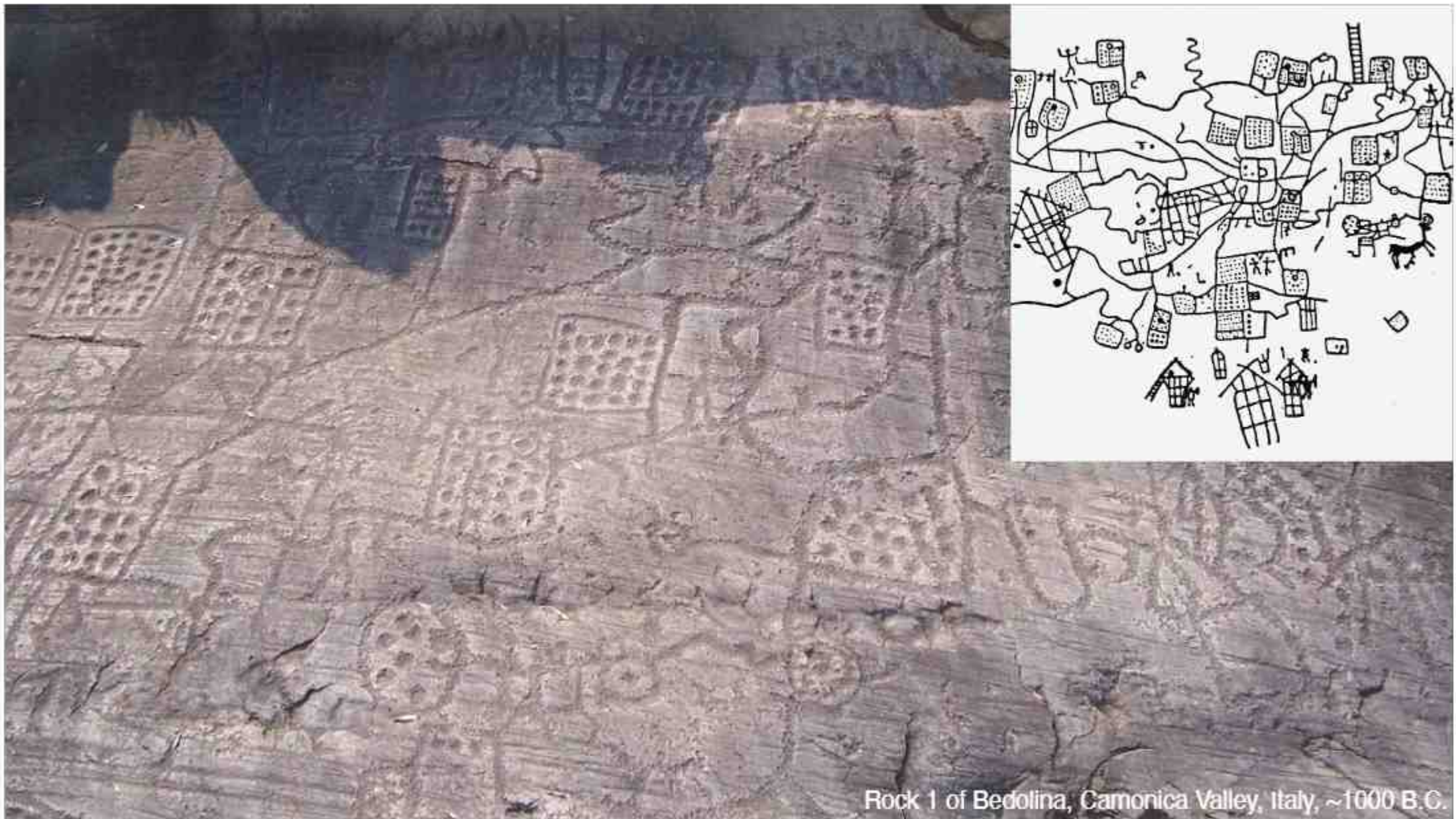


What is a map?

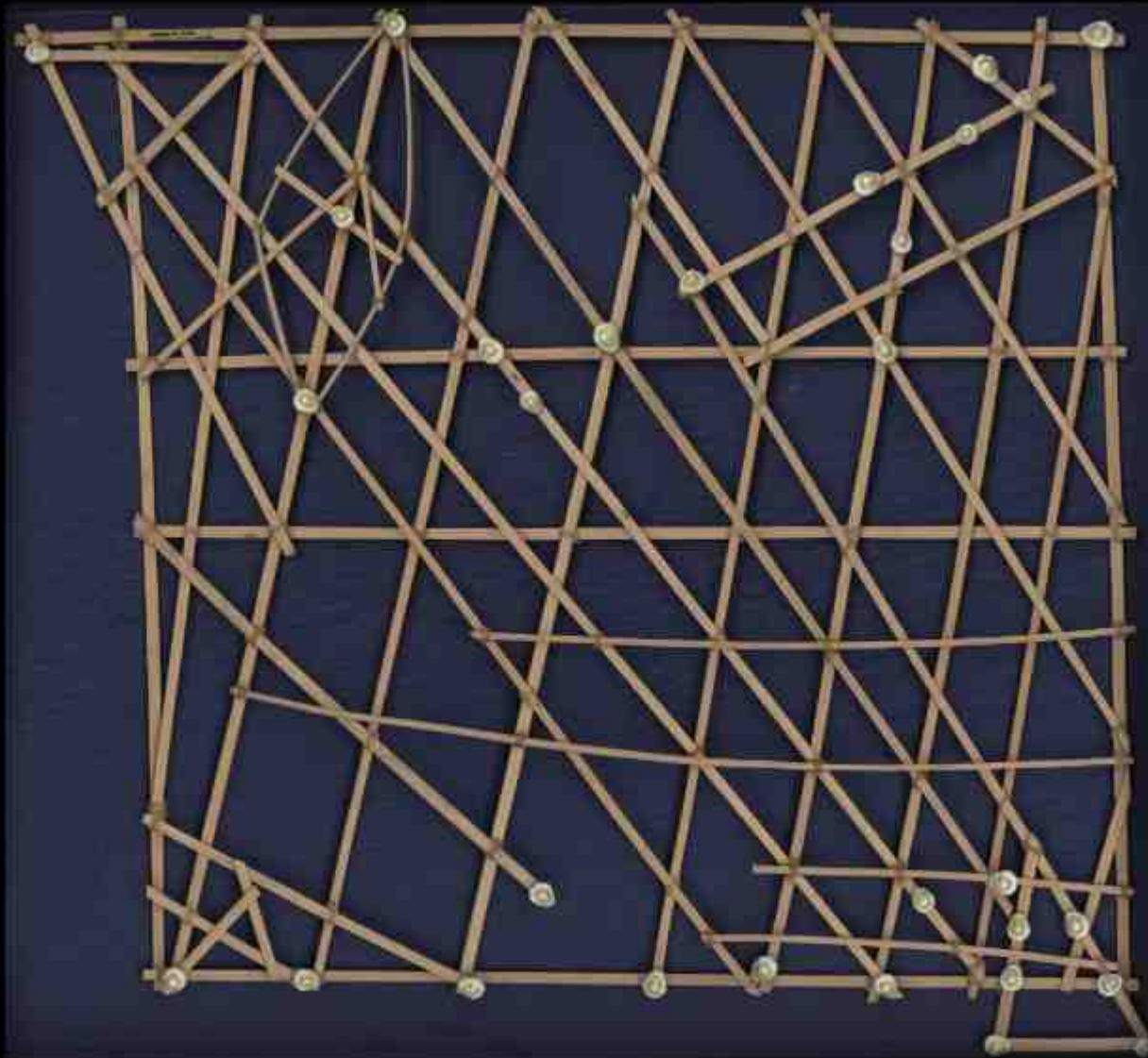
Or, are these maps?



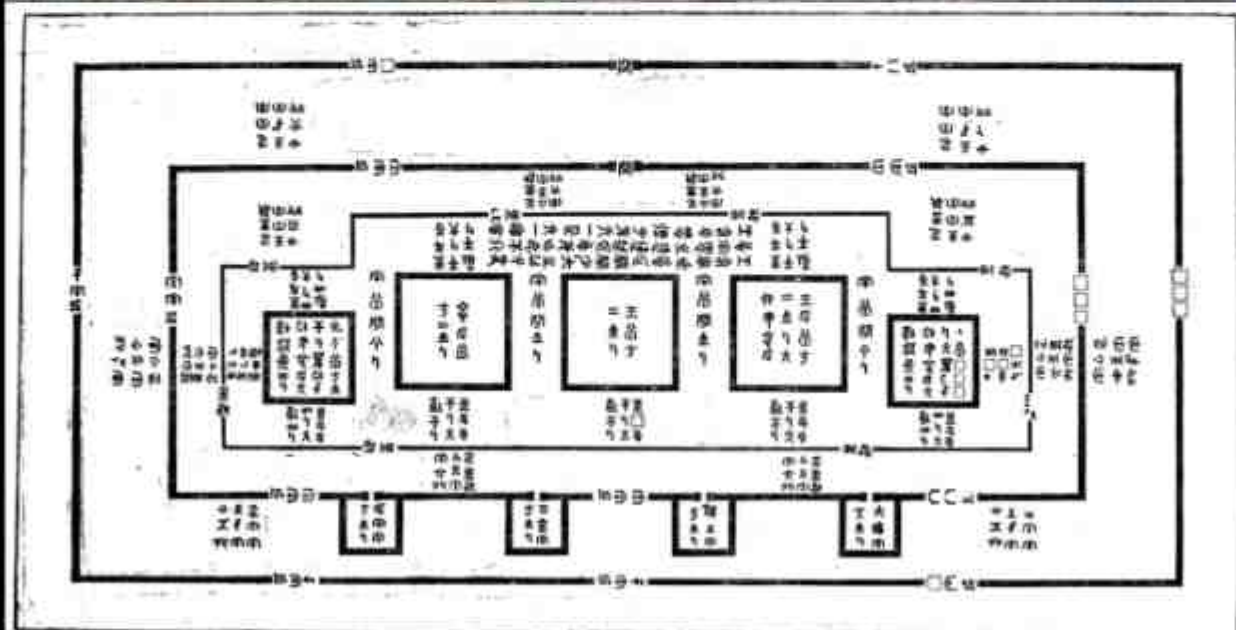
Çatalhöyük, Turkey, ~6000 B.C.



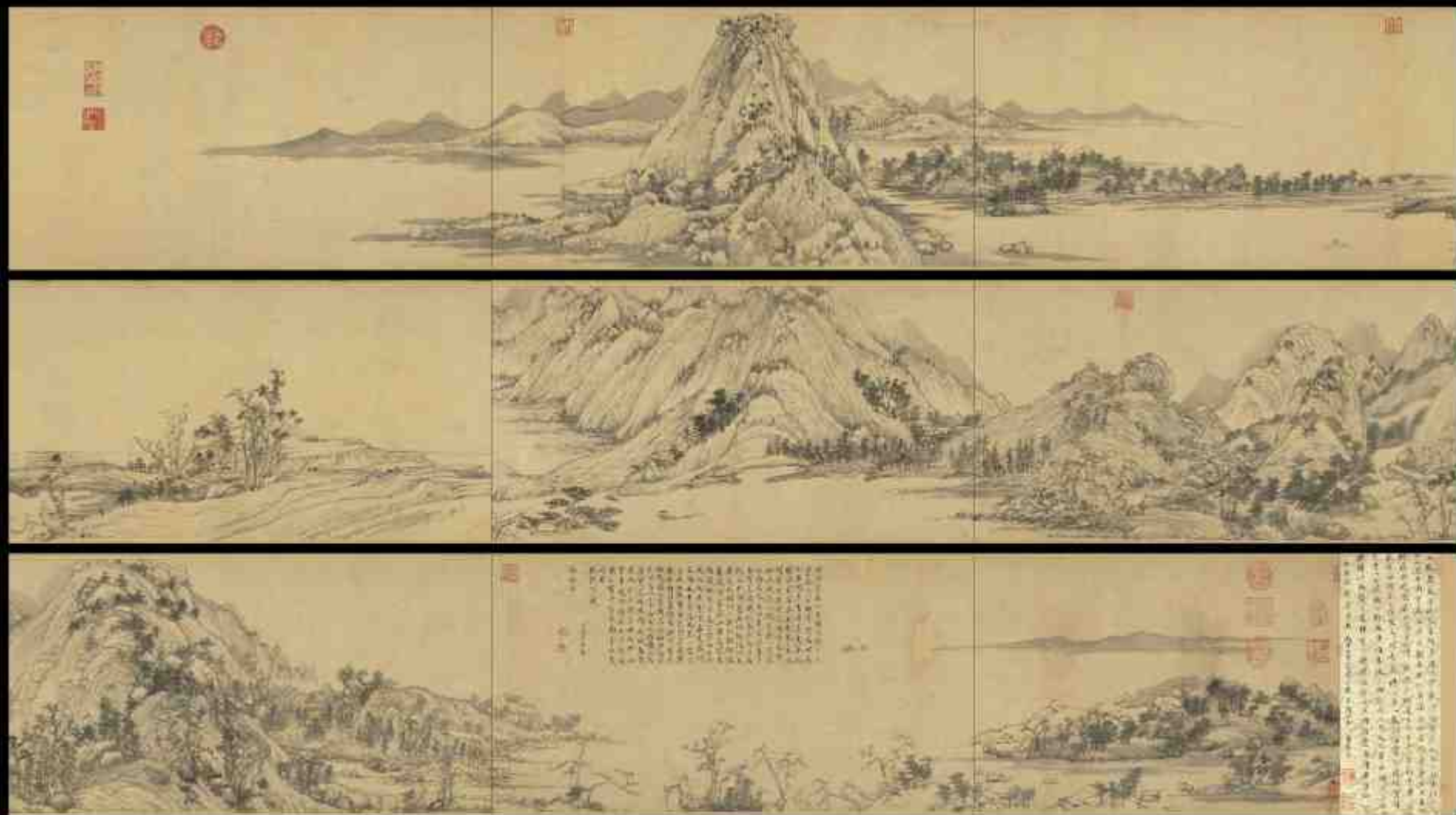
Rock 1 of Bedolina, Camonica Valley, Italy, ~1000 B.C.



Marshall Islands Stick Chart



《兆域圖》· 公元前三世紀
戰國中山王墓



《富春山居圖》無用師卷
十四世紀·黃公望





《始政四十周年紀念臺北市鳥瞰圖》・1935・吉田初三郎

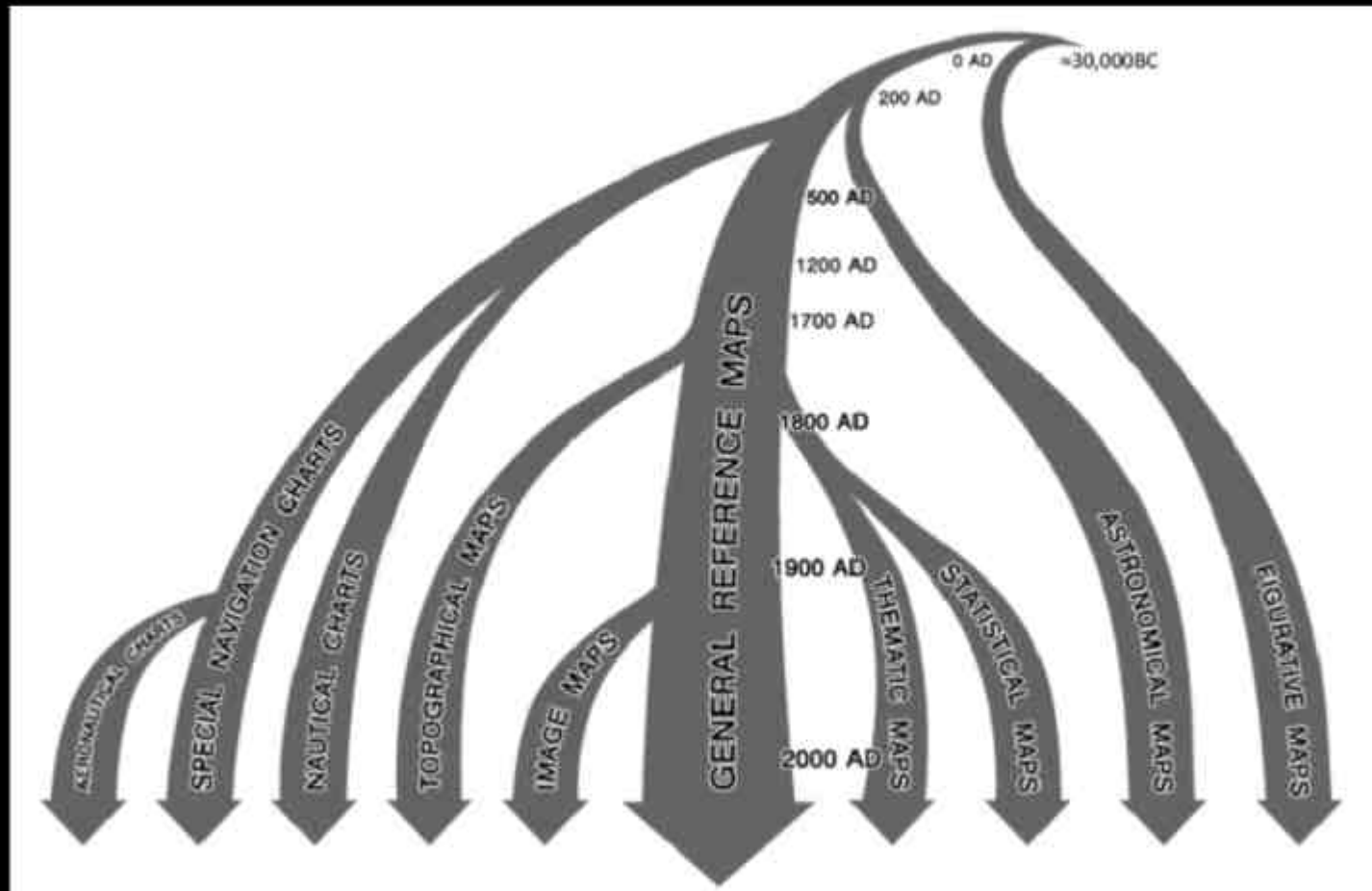
Maps and Cartography

- A map is the graphic representation of the geographical setting.
- Cartography is the making and study of maps in all their aspects.
- The most fundamental function of maps is to bring things into view.
- All maps have one thing in common: to add the geographic understanding of the viewer.
- All geographical maps are reductions. Thus, they are smaller than the region they portrays.
- All maps are abstractions of reality.

Purposes Maps Serve

- To store geographic information in spatial format.
- To serve mobility and navigation needs.
- For analytical purposes involving measuring and computing.
- To summarize voluminous statistical data and to assist in spatial forecasting and spotted trends.
- To visualize what otherwise would be invisible.
- To stimulate spatial thinking.

Diverse Map Types Through Cartographic Revolution

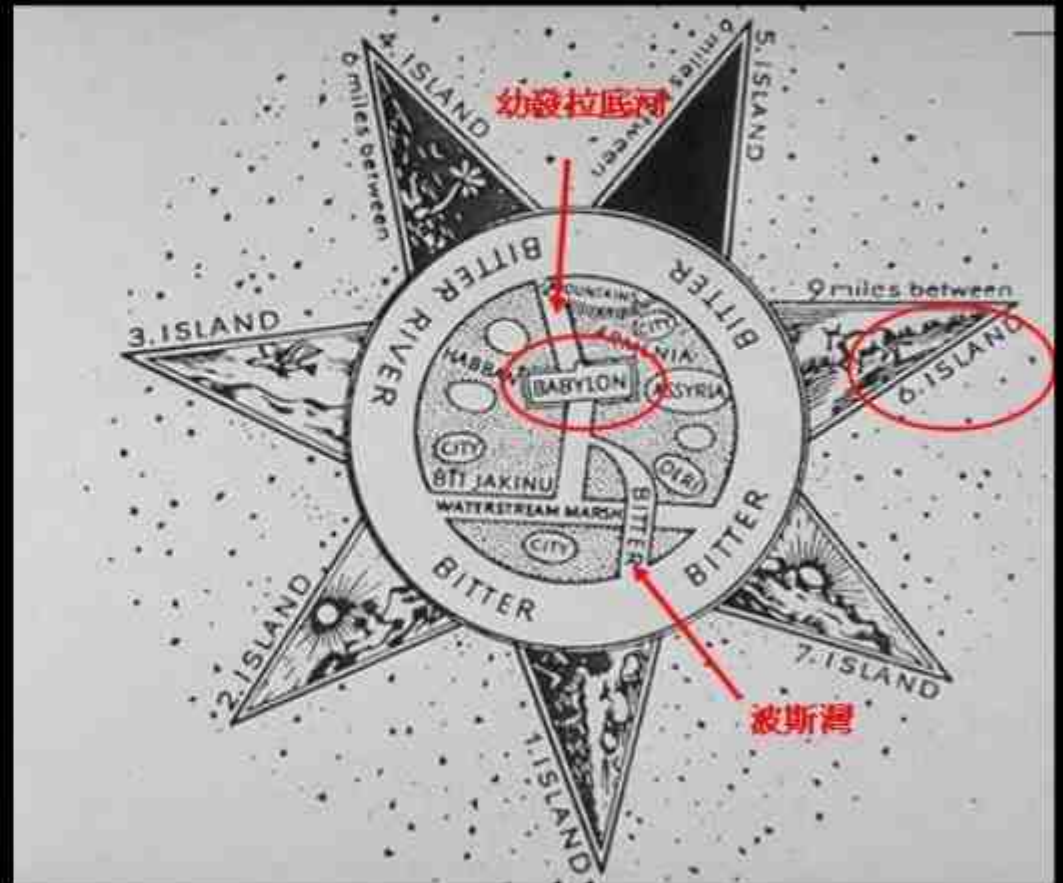


Robinson et al. (1995)

Ancient Maps



Babylonian Map of the World, ~800 B.C.



Greco-Roman World Maps



Ptolemy's World Map, 13th Century

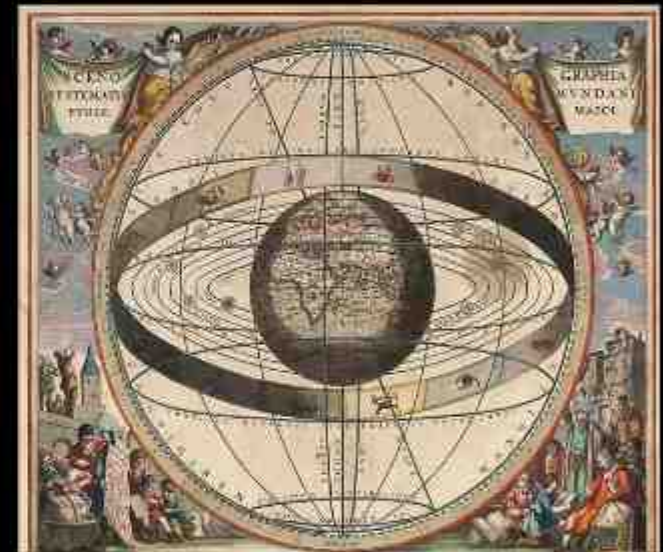


Claudius Ptolemy (托勒密) (c. 100 - 160 AD, Father of Geography)

Ptolemy's World Map



Ptolemy World Map, 1482, Nicolaus Germanus



世界最古老印刷地圖集拍出2 1 3 · 6 萬英鎊

2006年10月12日 07:50:52 來源：新華網

【字號 大 中 小】

【我要打印】

【我要糾錯】

【Email推薦： 提交】

新華網北京1 0月1 1日專電一冊世界最古老的印刷地圖集1 0日在倫敦索思比拍賣行以2 1 3 · 6 萬英鎊的價格成交，成為有史以來拍賣成交價最高的地圖集。

這本6 1頁的地圖集有5 2 9年歷史，原是英國沃丁頓勳爵的藏品，此次被一位私人收藏家買走。地圖集依據的是公元2世紀的地理學家克勞迪烏斯·托勒密的測繪成果。托勒密死後1 0 0 0多年間，他所制作的地圖一直被用作地理教學的標準資料。直到公元1 6世紀佛蘭芒人赫拉爾杜斯·墨卡托繪制了新的地圖，托勒密版本地圖“一統天下”的局面才宣告結束。

1 4 7 7年，在托勒密版本地圖集的基礎上，世界首本印刷地圖集在意大利的博洛尼亞出版。1 5年後，克裏斯托夫·哥倫布才從大西洋向美洲進發，開始環球航行。據拍賣行介紹，目前世上僅存兩本這種古老的印刷地圖集，都被私人收藏。

據悉，此前地圖集拍賣的成交價紀錄是1 4 6 · 4 萬英鎊。

[illegible]

The Universal Cosmography according to the Tradition of Ptolemy and the Discoveries of Amerigo Vespucci and others, 1507, Martin Waldseemüller

Church/Religious Maps



T and O Map

The T represents the Nile and the Don Rivers and the Mediterranean Sea that separates Europe, Asia, and Africa. The three continents were divided among the three sons of Noah.



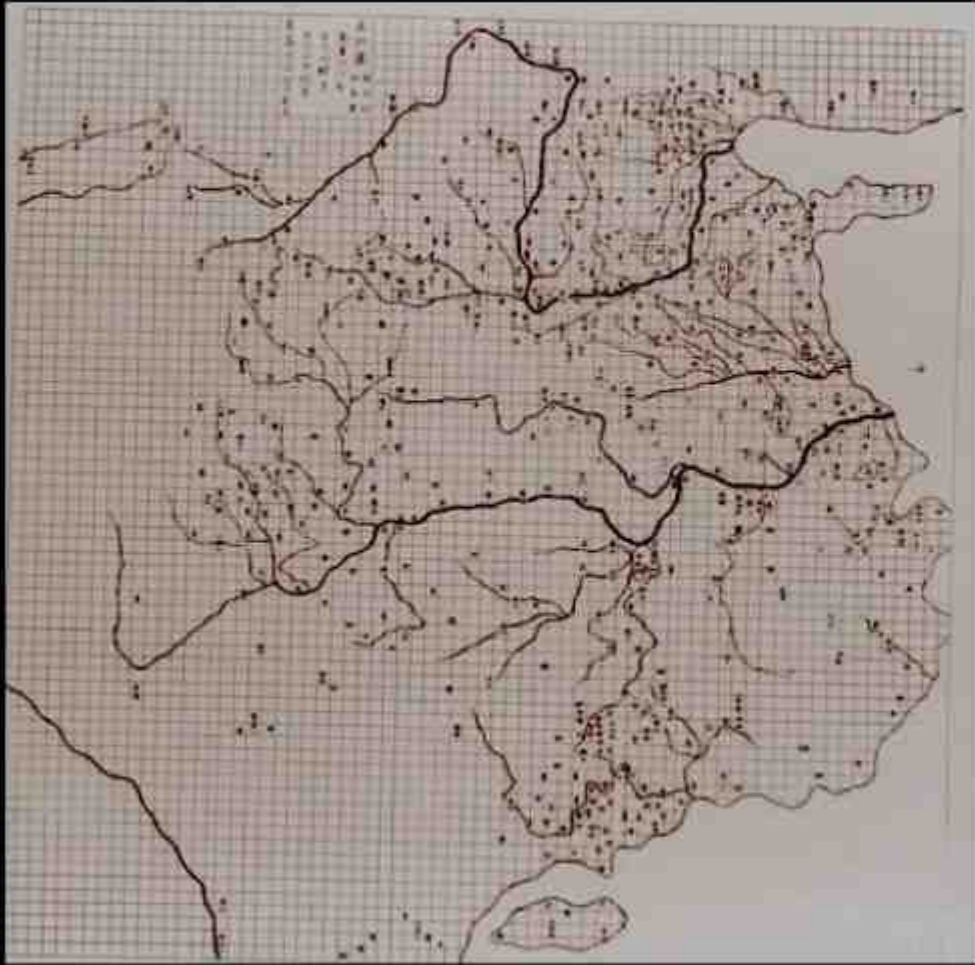
Hereford Mappa Mundi, c. 1300 AD

Ancient Islamic World Maps



World Map, 1154, al-Idrisi

Ancient Chinese Maps



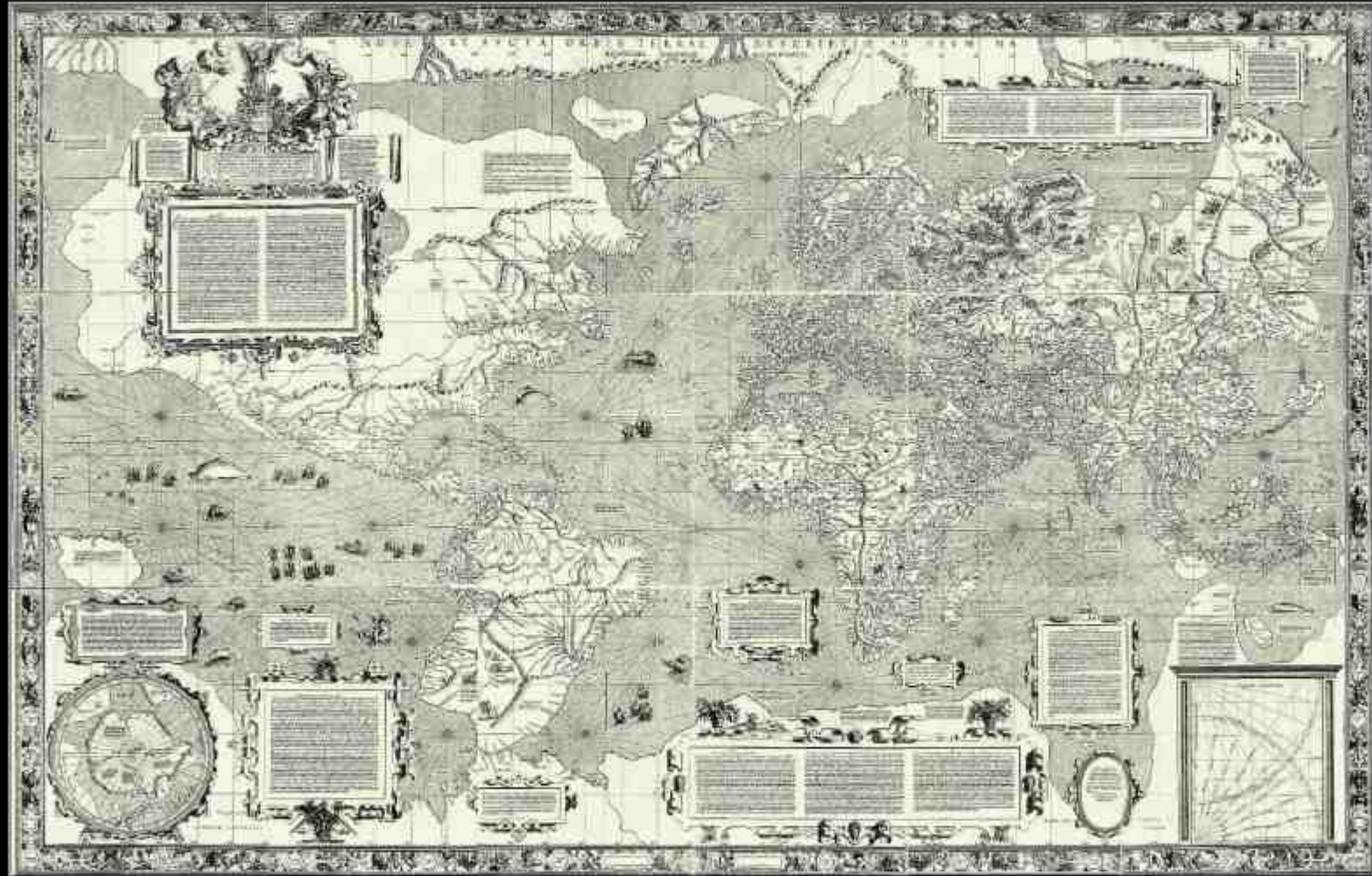
《禹跡圖》· 1137

Yu was the Chinese hydraulic engineer who was commissioned by the emperor to solve flooding problems and insure Chinese citizens' safety. He traveled all over China in studying flooding hazards and made a very detailed map of major rivers and their tributaries. The use of a rectangular grid system as a map scale helped Yu in keeping spatial records. The detail and accuracy of this map is superior to the contemporary T & O maps of Europe.



漢代長沙國南部地形圖，公元前二世紀，長沙馬王堆三號墓

16-17th Century: Maps after the Age of Discovery



Mercator World Map, 1569, Gerardus Mercator

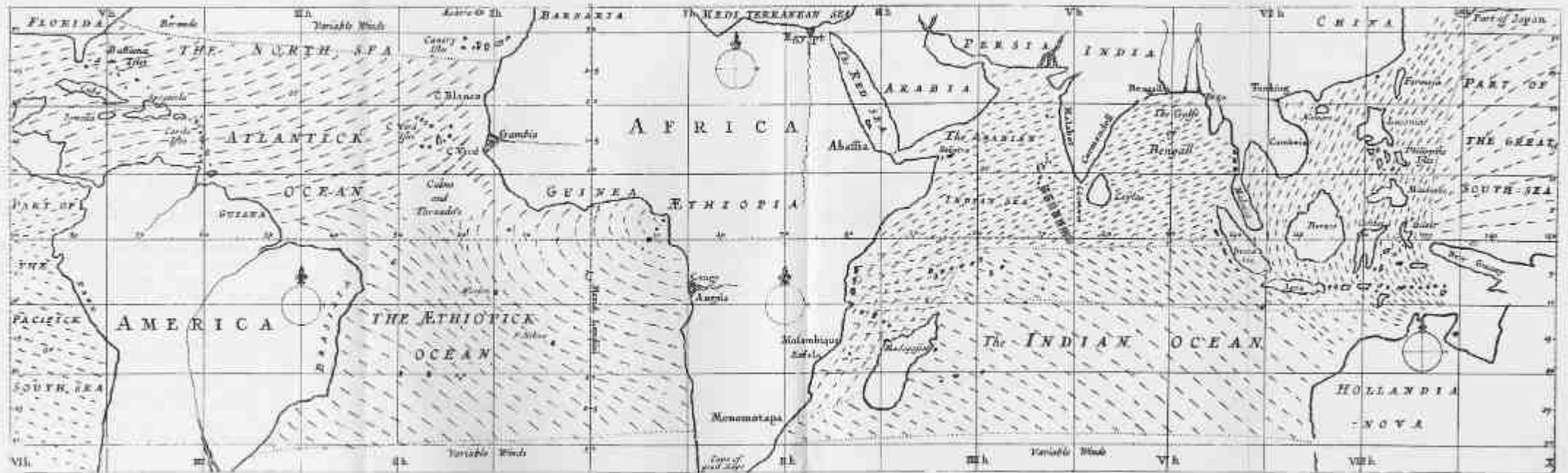


British Isles Map, 1646, Willem Blaeu

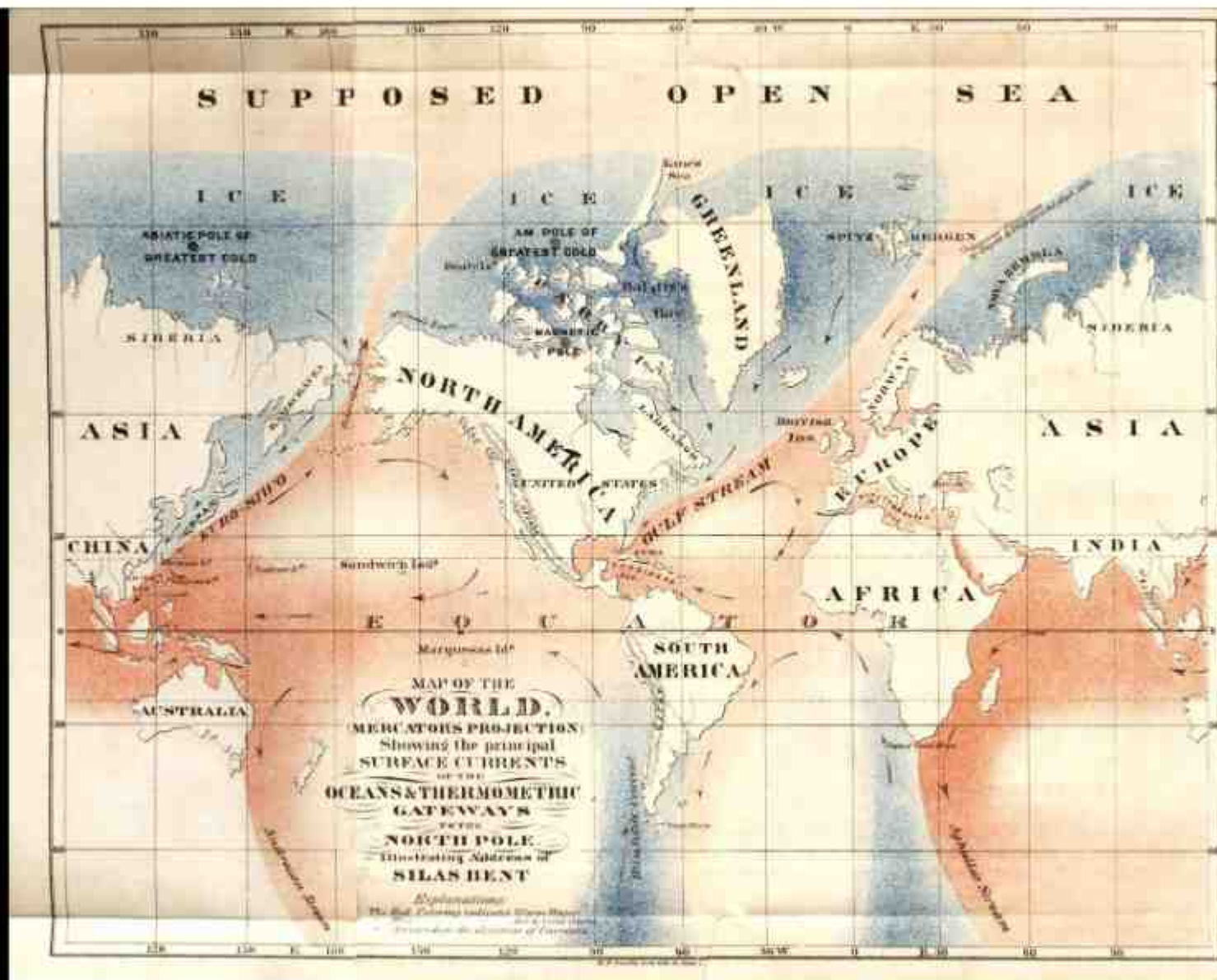


Willem Janszoon Blaeu
1571 – 1638
Official cartographer for the East
India Company

17-18th Century: Thematic Maps



Edmond Halley made a world map of trade winds and monsoons (first meteorologic map) in 1686.



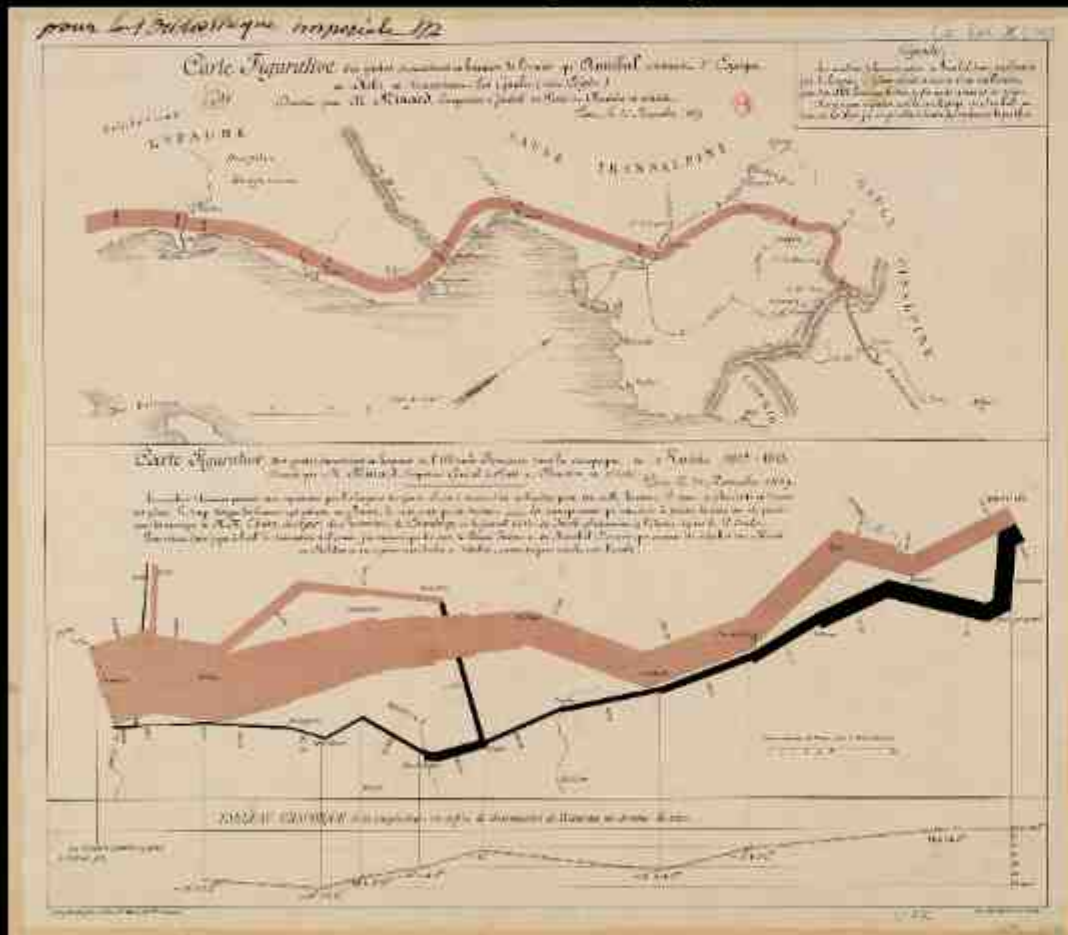
Oceans and Thermometric Gateways to the North Pole, 1872, Silas Bent

“Best Graphic Ever Produced”

Edward Tufte

“The Visual Display of Quantitative Information”

Hannibal's famous military campaign of 218 BC



Napoleon's 1812 march on Moscow

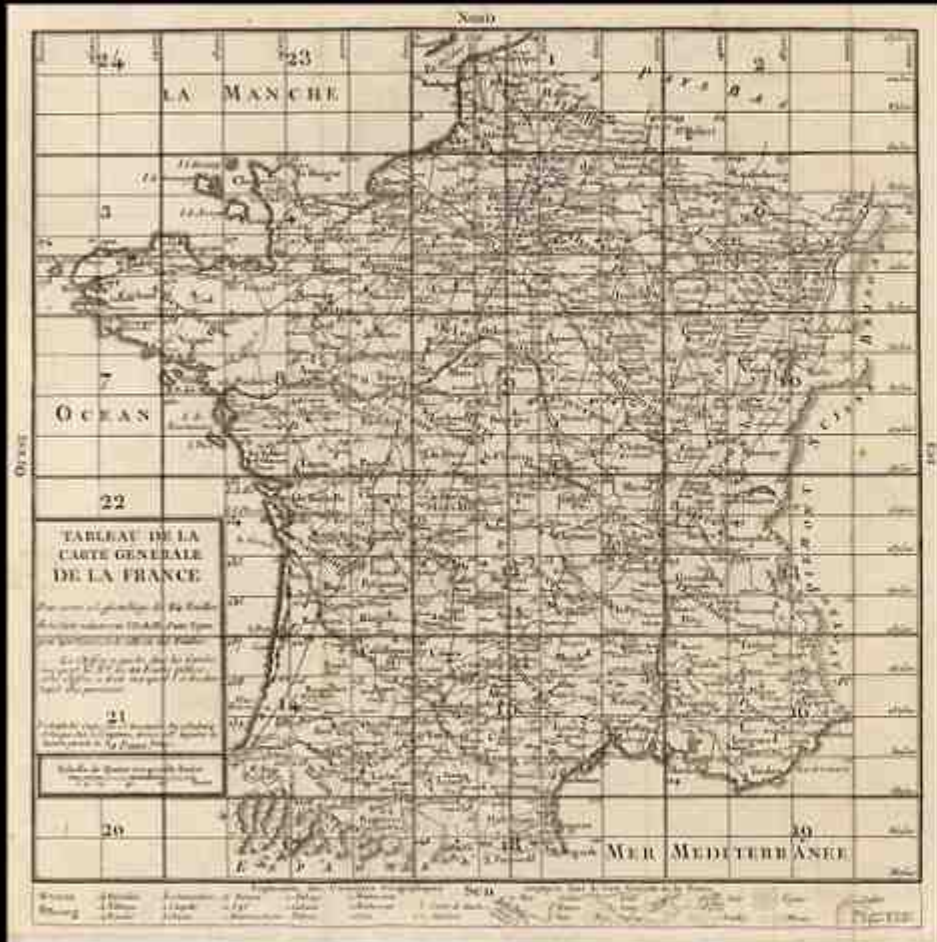


Charles Joseph Minard
1781-1870

French civil engineer

The tremendous numbers of casualties suffered by both armies are shown by the thinning of the lines (1 millimeter of thickness is equal to 10,000 men), which charts represent their routes through space and time.

18th Century: Survey Maps



Carte de Cassini, Cassini Map or Academy's Map



the pioneering topographic and geometric survey of the entire Kingdom of France, created over four generations of the Cassini family from 1750 to 1815 at a uniform scale of 1:86,400 and comprising 182 engraved sheets that could be assembled into a massive wall map approximately 39 feet high by 38 feet wide.

Geographic extent of German city of Dusseldorf at different time periods



1887



1900

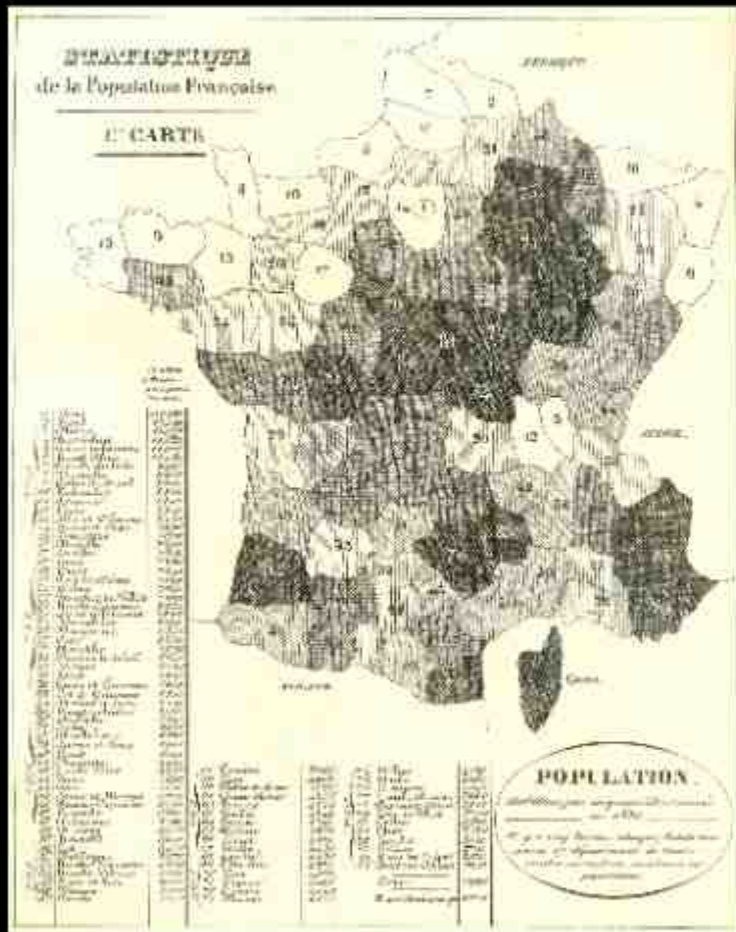


1905

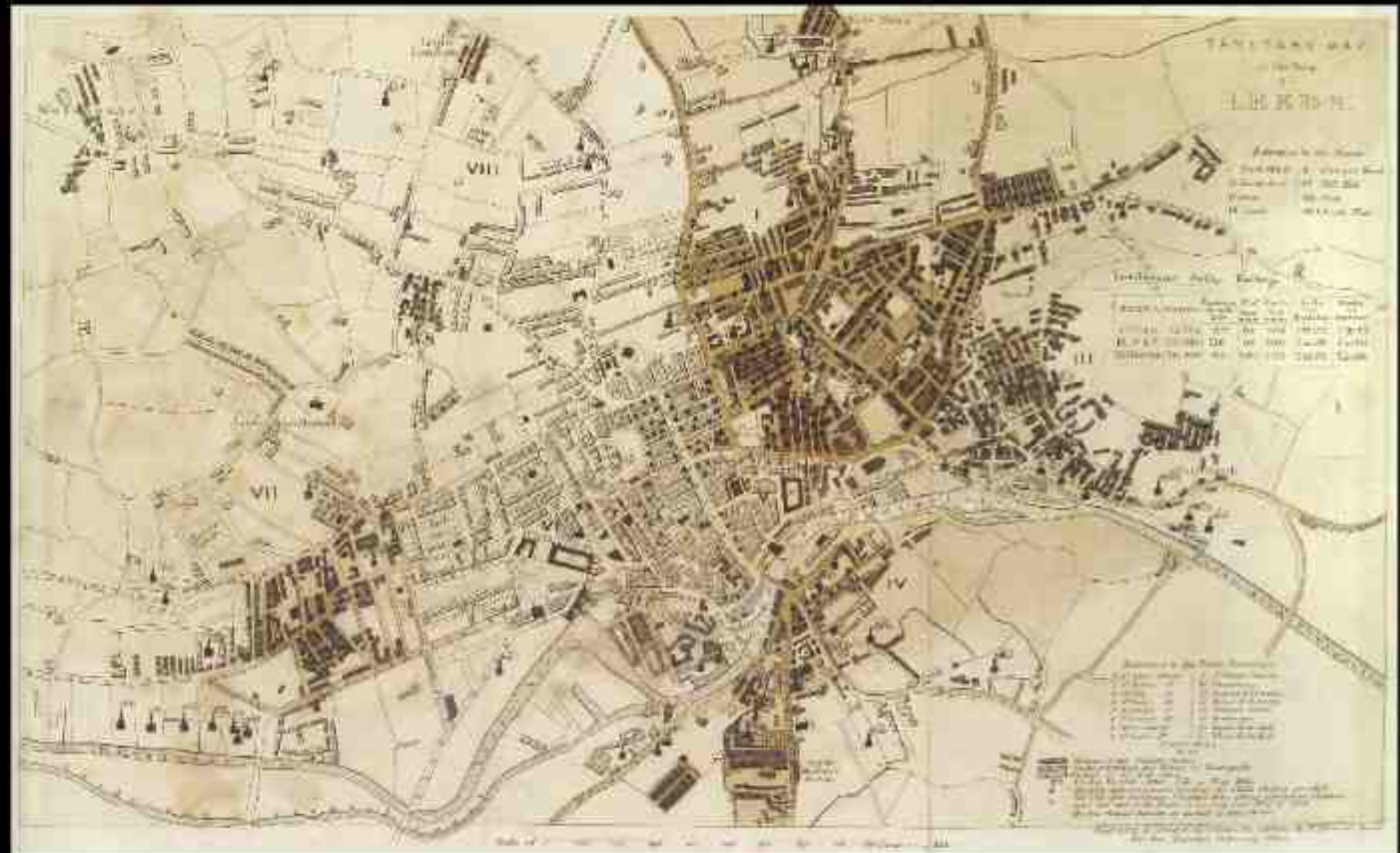


1910

19th Century: Maps about Population and Society



Statistique de la Population Française,
1836, Adolphe d'Angeville



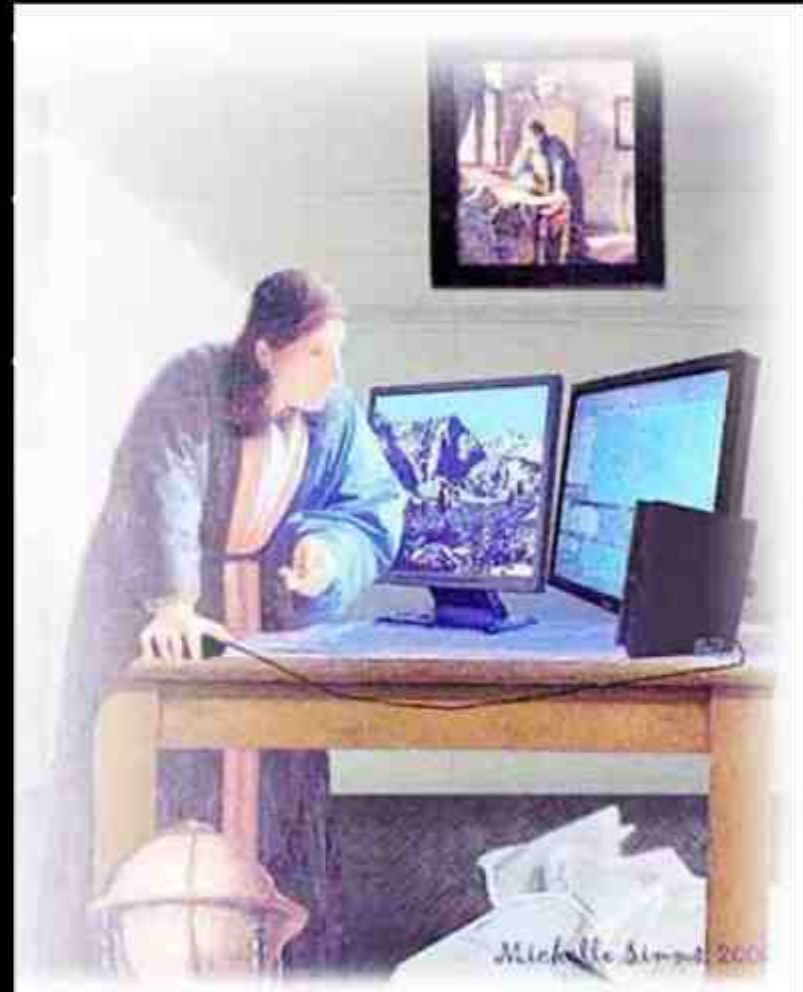
Sanitary Map of Leeds, 1832, Edwin Chadwick

Disadvantages of Creating Printed Maps

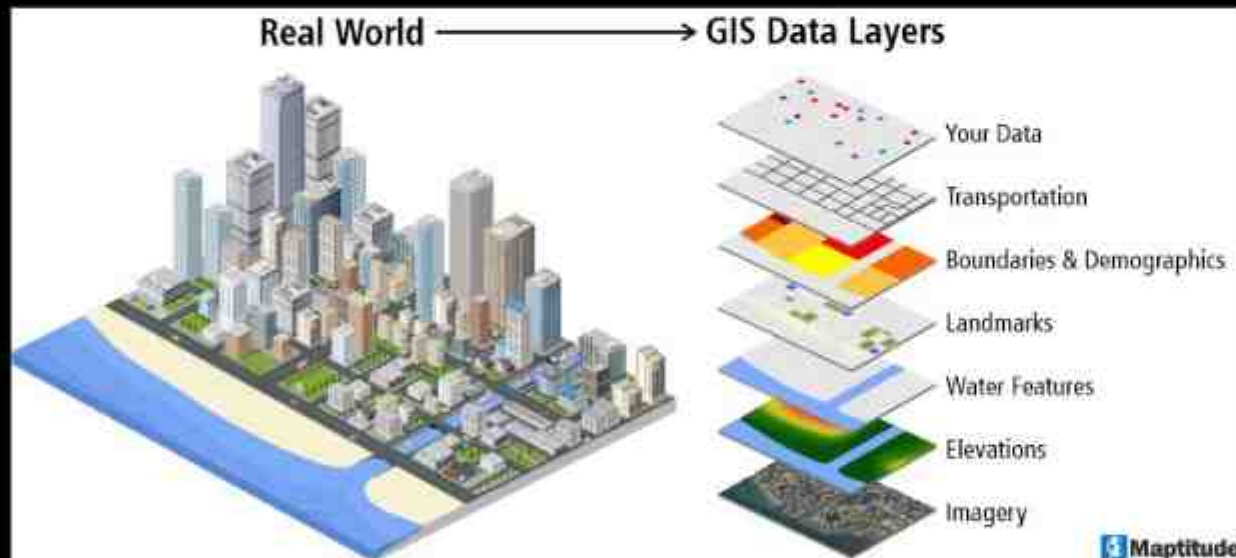


- Time-consuming
- Data is not reusable
- Lack of flexibility to change the content

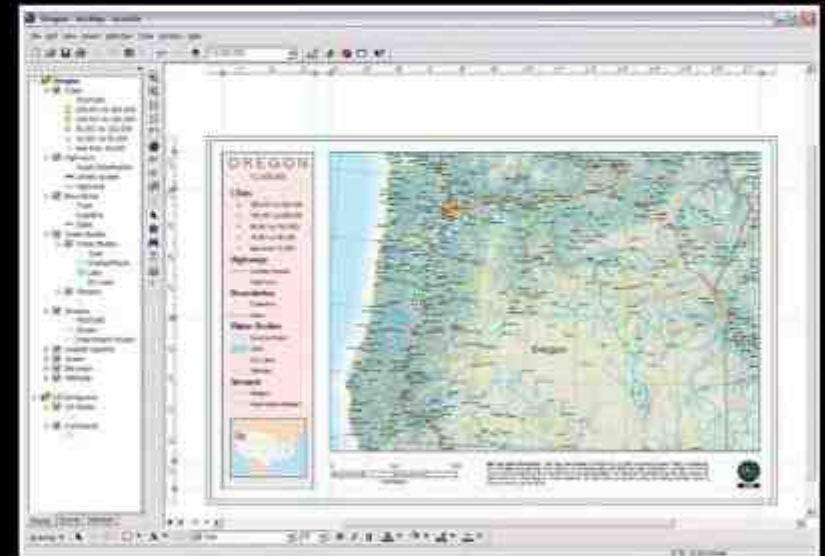
Disadvantages of Creating Printed Maps



20th Century: Geographic Information System

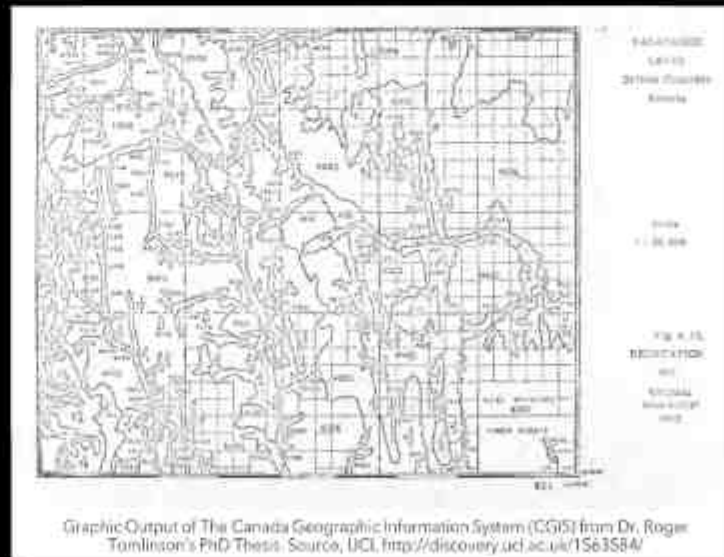


<https://www.caliper.com/maptitude/blog/what-are-the-best-gis-data-sources/default.htm>



The Emergence of GIS

- GIS = GI system = Geographic Information System
- The definition of GIS
 - GIS is fundamentally about the use of digital data to represent space and time (Longley et al., 2005).
- Roger Tomlinson (Father of GIS) developed first computerized GIS, Canadian Geographic Information System (CGIS) in 1963.



Advantages of Learning GIS

Mapping without GIS

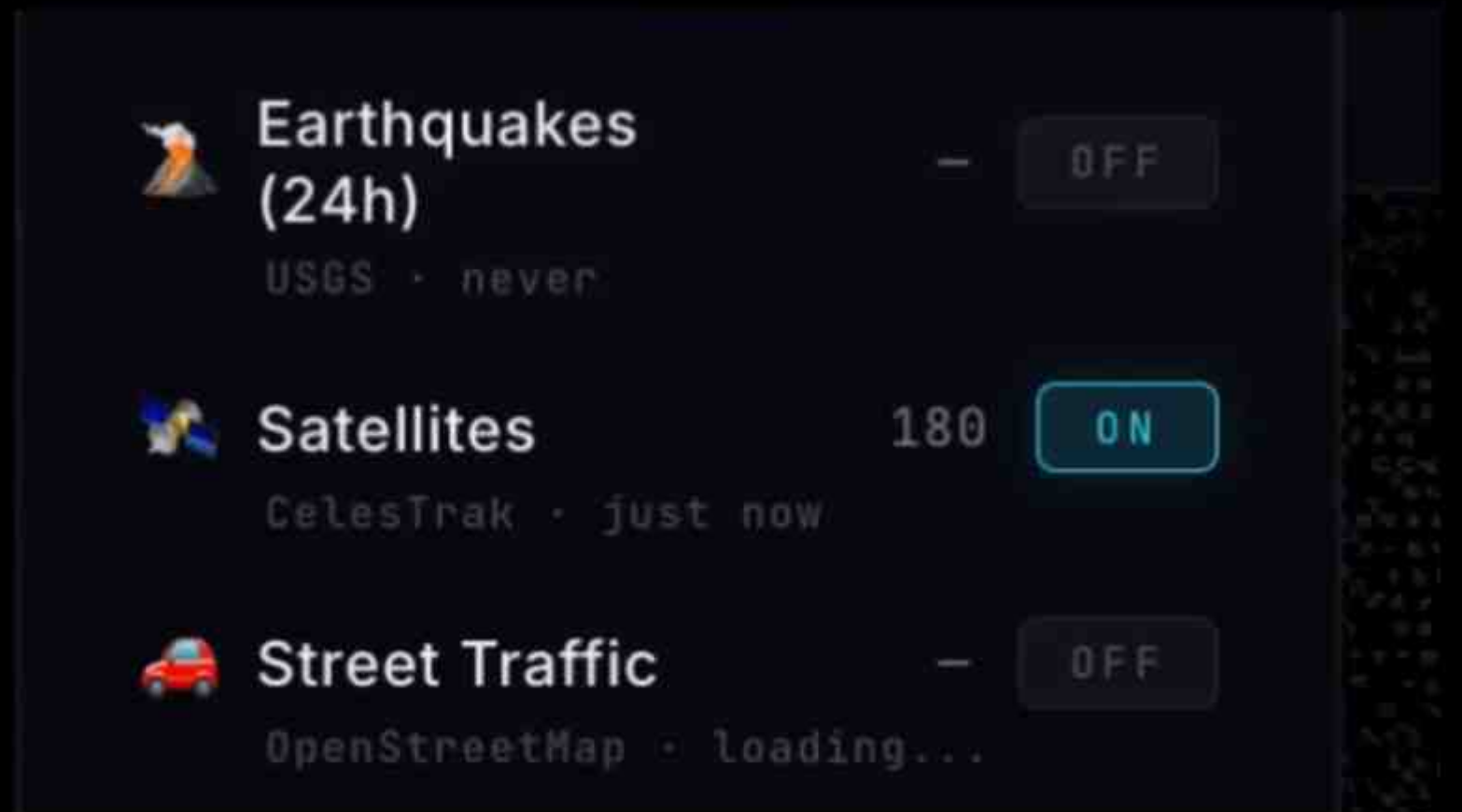
- Time-consuming
- Data is not reusable
- Lack of flexibility to change the content

Use GIS for mapping

- Create a map instantly
- Data is reusable
- High flexibility to change the content

21st Century: Web Maps

- Interactive
- More than 2D
- Link to devices
- Real-time
- Big data
- Crowd-source



Location-Based Service (LBS)

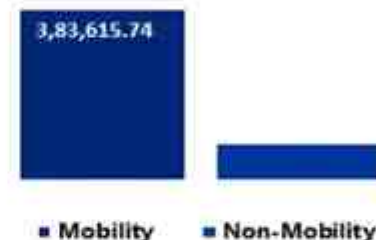
LBS (Location Based Services) Market Overview



Global Market Analysis, By Region, 2035

North America is anticipated to dominate the market from 2026 to 2035, owing to increasing smartphone adoption, growing demand for location-based applications, and a rising need for real-time location data across various industries.

Global Market Analysis, By Type (in USD million) 2035



Key Players in the Market

- ALE International
- Apple INC
- Cisco Systems, Inc.
- Ericsson group
- Esri
- GL Communications Inc
- Glympse Inc.
- HERE
- Hewlett Packard Enterprise Development LP
- HID Global Corporation
- IBM
- IndoorAtlas

Growth Rate



Growth Drivers



- Emerging Technologies for Location-Based Services
- Rise in 5G Network Rollout
- Growth of E-commerce and On-demand Services

Challenges



- Privacy and Data Security Concerns
- High Cost of Infrastructure and Implementation

Geospatial Artificial Intelligence (GeoAI)



Model the real world for prediction

Aerial imagery is used to extract imagery of buildings and roads in Grenada to identify the population and infrastructure at risk for landslides.

DEFINITION

What is GeoAI?

Geospatial artificial intelligence (GeoAI) is the application of artificial intelligence (AI) fused with geospatial data, science, and technology to accelerate real-world understanding of business opportunities, environmental impacts, and operational risks. Organizations are modernizing operations to run at scale through automated data generation and approachable spatial tools and algorithms.

01. Extract rich geospatial data with deep learning

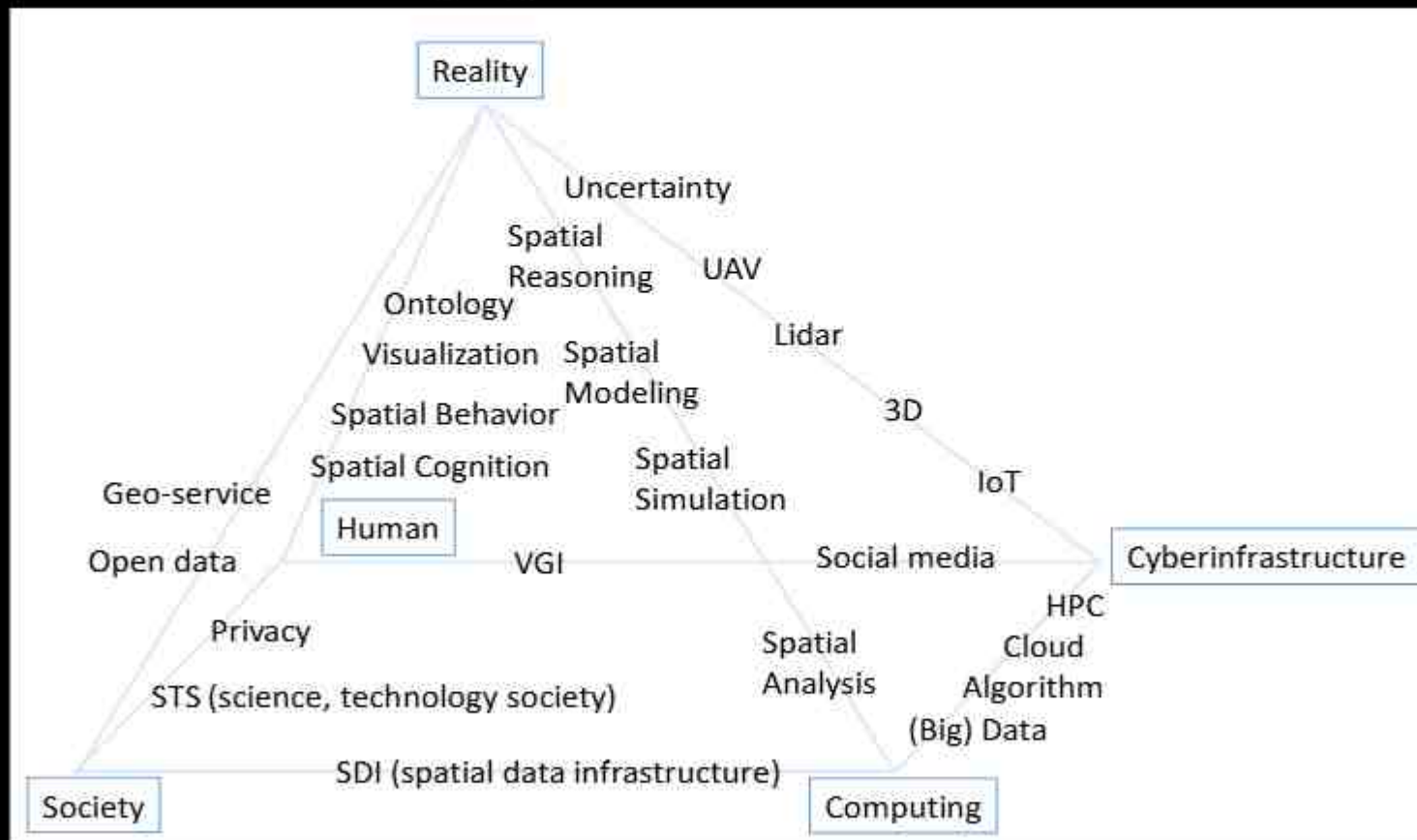
Save time by automating the extraction, classification, and detection of information from data such as imagery, video, point clouds, and text.

02. Perform predictive analysis using machine learning

Build more accurate models. Detect clusters, calculate change, find patterns, and forecast outcomes with spatial algorithms backed by experts.

New Era of GIS

- GIS = GI Science = Geographic Information Science



Policy paper

UK Geospatial Strategy 2030

Updated 3 August 2023

This was published under the 2022 to 2024 Sunak Conservative government

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[Mission 1: Embrace enabling technologies to accelerate geospatial innovation](#)

[Mission 2: Drive greater use of geospatial applications and insights across the economy](#)

[Mission 3: Build confidence in the future geospatial ecosystem](#)

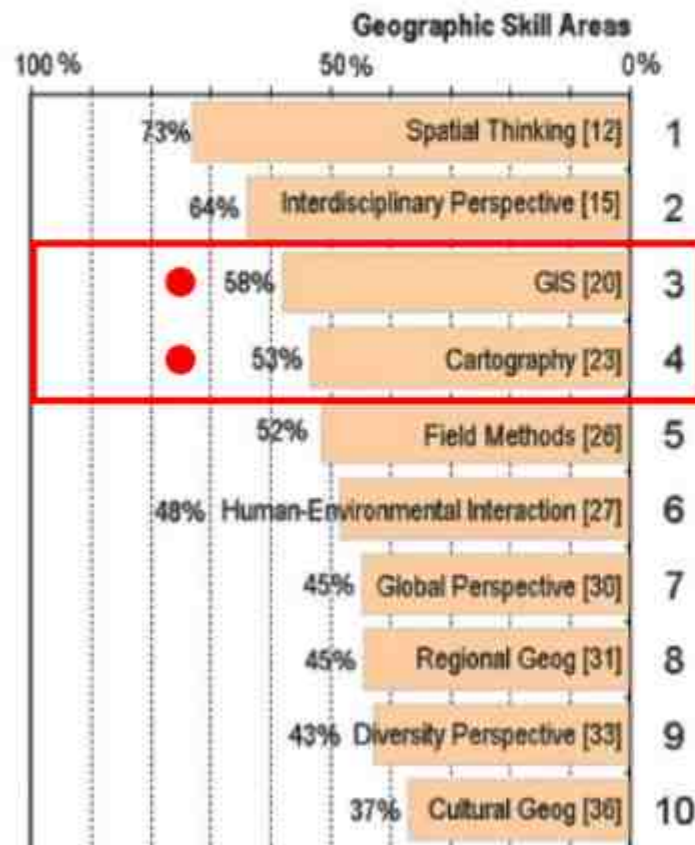
About the Geospatial Commission

The Geospatial Commission was established in 2018 by the government as an expert committee responsible for setting the UK's geospatial strategy and coordinating public sector geospatial activity. We are part of the Department for Science, Innovation and Technology, playing a crucial role to deliver improved public services, create new better-paid jobs and grow the economy.

Our aim is to unlock the significant economic, social and environmental opportunities offered by location data, applications and services and to advance the UK's global geospatial expertise. The Geospatial Commission has a mandate to research, initiate, drive and deliver changes by working in partnership with others. This means we:

- Provide strategic oversight of the geospatial ecosystem in the UK, setting geospatial strategy, policy and standards
- Hold the budget for the public sector's largest investment in location data, the Public Sector Geospatial Agreement (PSGA)
- Make targeted investments in projects that accelerate innovation and adoption of location data, applications and services

Professional Knowledge in Geography: Employment and Entrepreneurship Trends



Skills in Professional Geography: An Assessment of Workforce Needs and Expectations*

Michael Solem
Association of American Geographers

Ivan Cheung
Insurance Institute for Highway Safety

M. Beth Schlemper
University of Toledo

Sector	Alumni (N = 280)	Employer (N = 447)	AAG contact database (N = 4,996)
Higher education	52%	52%	68%
Government	22%	30%	14%
For-profit company	17%	15%	13%
Nonprofit organization	4%	3%	2%
Unclassified	5%	0%	3%

地理資訊相關課程



GIS Jobs

- 地理資訊公司：瑞竣、松旭、群璇等
- 電子地圖公司：Garmin、Here、TomTom、勤崴等
- 空間資料應用：Google、Apple、Microsoft、Amazon等
- 衛星與遙測：NASA、ESA、JAXA、太空中心、太遙中心等
- 智慧產業公司：Acer、遠傳、富鴻網等
- 地理資訊大數據：中華電信、遠傳電信、銀行、房產、保險等
- 工程顧問公司：中興社、中興工程、世曦工程等
- 防災產業：災防中心、國震心中、興創知能、北極星等
- 無人機公司：翔隆、智飛、經緯航太等
- 測量公司：中興測量、自強工程、迅聯、詮華等
- 測量儀器公司：Trimble、Leica、台儀、宏遠等
- 資源調查與規劃機關：水利署、地礦中心、林保署、營建署、都發局等
- 測量製圖公務機關：國土測繪中心、地政事務所等

Geospatial Startups

Top 14 Geospatial Startups in 2024

Updated: February 9, 2024 | [Startups](#)

Think Geospatial is boring and everything's been done before?

Think again.

If you're in the Geospatial industry or are interested in starting a company in this space, this article dives into 14 of the hottest fast-growing Geospatial startups.

14 Geospatial Startups

- 1) Setologic
- 2) Propeller Aero
- 3) Kayrros
- 4) Mapillary
- 5) Lattitude40
- 6) Ecopra.AI
- 7) TerraMagna
- 8) GeoQuic
- 9) QGIS.com Spatial Intelligence
- 10) Hydrocat
- 11) Airspace Link
- 12) Descartes Labs
- 13) VETRO FiberMap
- 14) Planet

14 Geospatial Startups

The Complete Unicorn List

Company	Valuation	Location
1	\$1.2B	London, UK
2	\$1.1B	London, UK
3	\$1.0B	London, UK
4	\$900M	London, UK
5	\$800M	London, UK
6	\$700M	London, UK
7	\$600M	London, UK
8	\$500M	London, UK
9	\$400M	London, UK
10	\$300M	London, UK
11	\$200M	London, UK
12	\$100M	London, UK
13	\$50M	London, UK
14	\$20M	London, UK

Sheet

1,016 Unicorns

Information about their valuation, HQ's location, founded year, name of founders, funding amount and number of employees.

Get the Sheet for \$50

The All-In-One Newsletter for Startup Founders

90% of startups fail. Learn how to not be with our weekly guides and stories. [Join 40,000+ founders.](#)

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國家考試 - 測量製圖

類科	普通科目	專業科目
高考測量製圖	●1.國文(作文80%、測驗20%) ※2.法學知識與英文(中華民國憲法20%、法學緒論20%、英文60%)	3.土地測量法規及實務(包括土地法、國土測繪法及地籍測量法規) 4.測量學(包括平面測量、大地測量及衛星定位測量) 5.誤差理論及實務 6.空間資訊理論及實務(包括航空測量與遙感探測、地圖學及地理資訊系統)
普考測量製圖	●1.國文(作文80%、測驗20%) ※2.法學知識與英文(中華民國憲法30%、法學緒論30%、英文40%)	3.測量學與土地測量法概要 4.誤差理論概要 5.空間資訊概要

